

Resilience Optimisation of Tipping-Prone Systems Through Spatial Heterogeneities

SIAM Conference on Nonlinear Waves and Coherent Structures (NWCS26)
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Aurora Faure Ragani
Robbin Bastiaansen

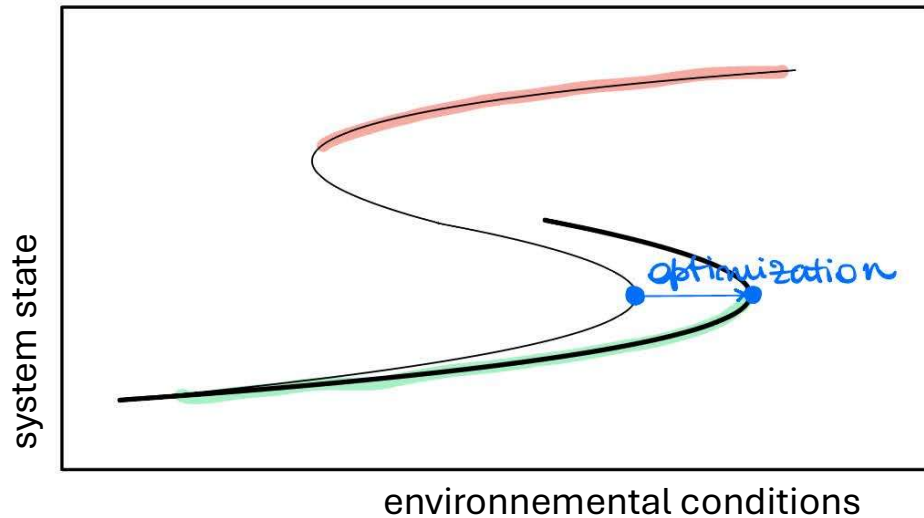
Idea: local interventions to prevent large-scale changes



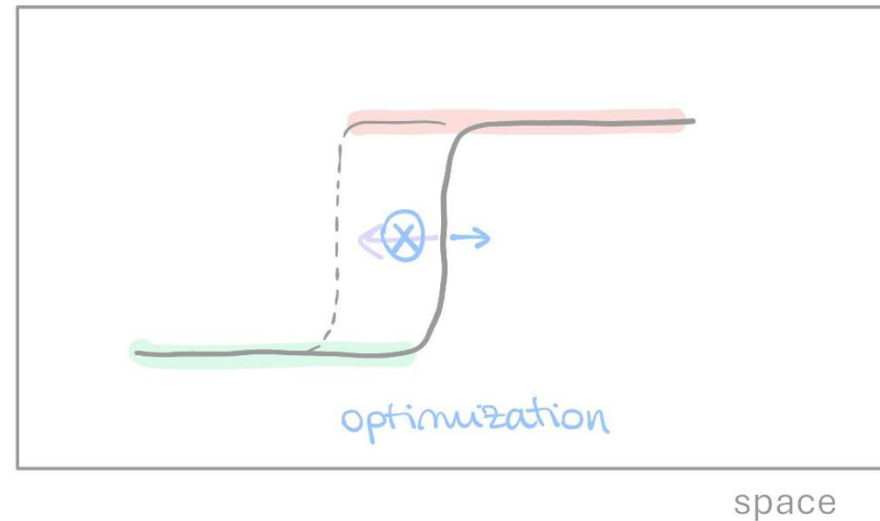
Andrew Millison – youtube (The Great Green Wall series)

Idea: optimise solutions by applying spatial heterogeneity

Bifurcation diagrams

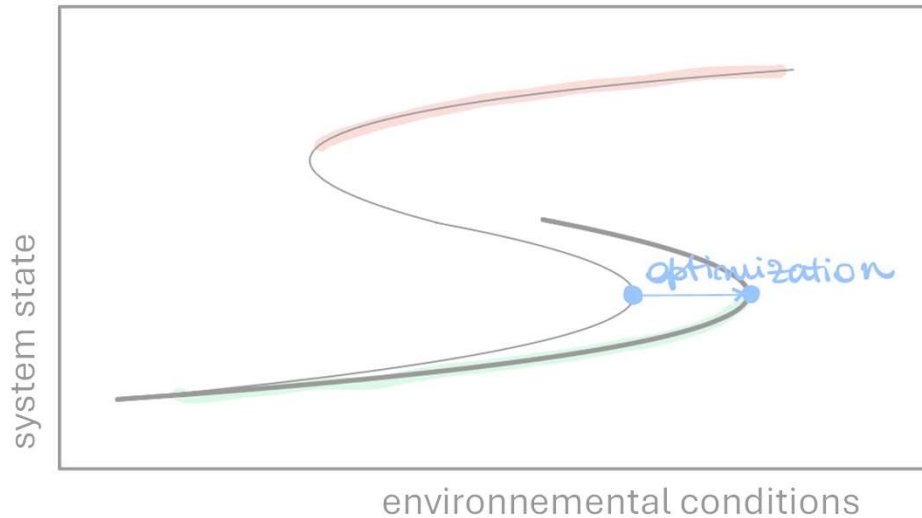


Front dynamics

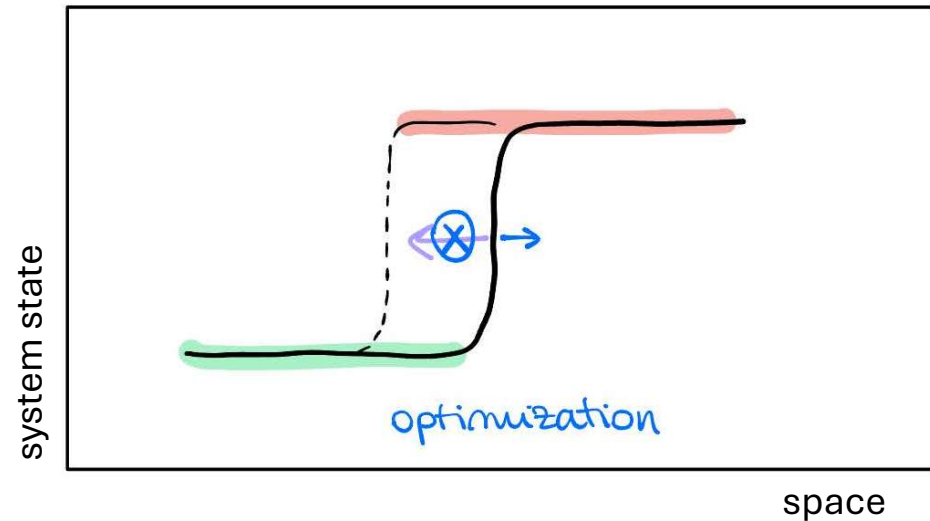


Idea: optimise solutions by applying spatial heterogeneity

Bifurcation diagrams



Front dynamics



Roadmap

1. Spatial tipping mechanisms

- Effect of spatial heterogeneity on bifurcation diagrams
- Effect of spatial heterogeneity on front dynamics

2. Optimisation framework

3. Application

- Equilibrium optimisation (acting on bifurcation diagrams)
- Final-state optimisation (acting on front dynamics)

Roadmap

1. Spatial tipping mechanisms

- Effect of spatial heterogeneity on bifurcation diagrams
- Effect of spatial heterogeneity on front dynamics

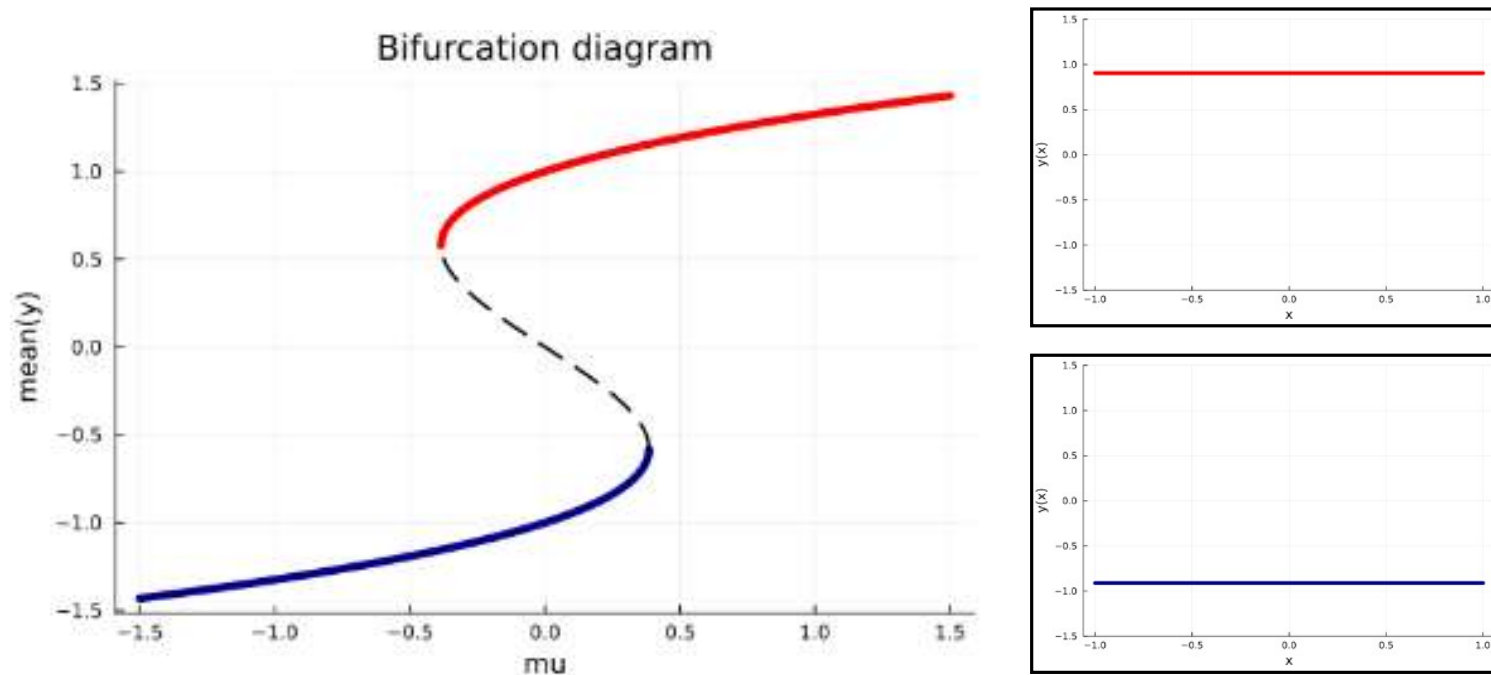
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Effect of spatial heterogeneity on bifurcation diagrams

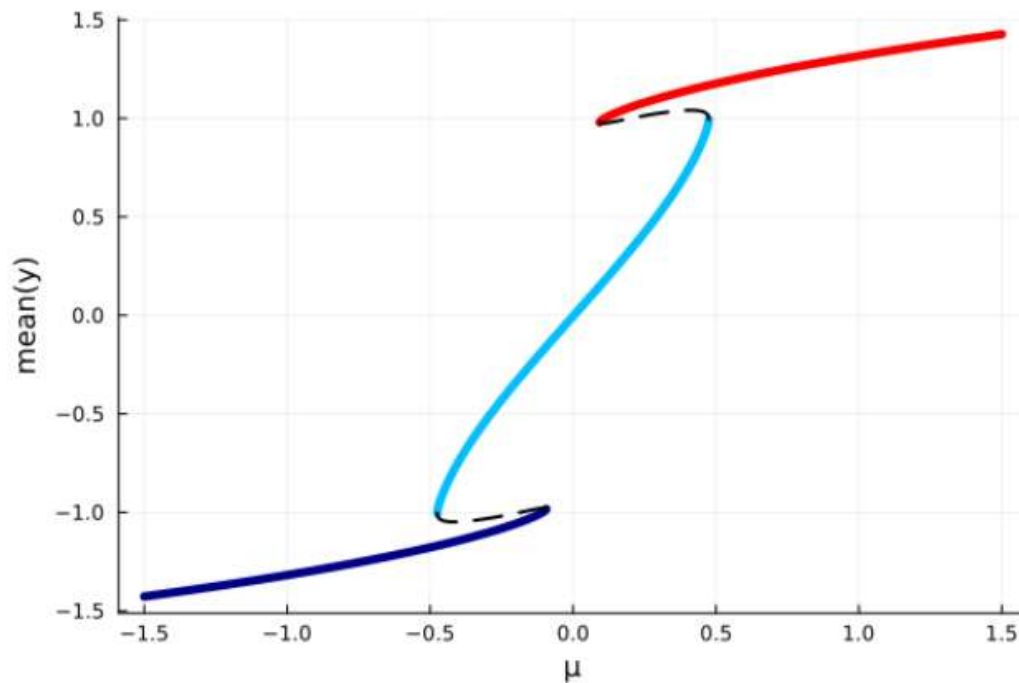
$$y_t = D y_{xx} - \partial V(y; \mu) / \partial y$$



Aurora Faure Ragani (a.faureragani@uu.nl)

Effect of spatial heterogeneity on bifurcation diagrams

$$y_t = D y_{xx} - \partial V(y, \mathbf{x}; \mu) / \partial y$$



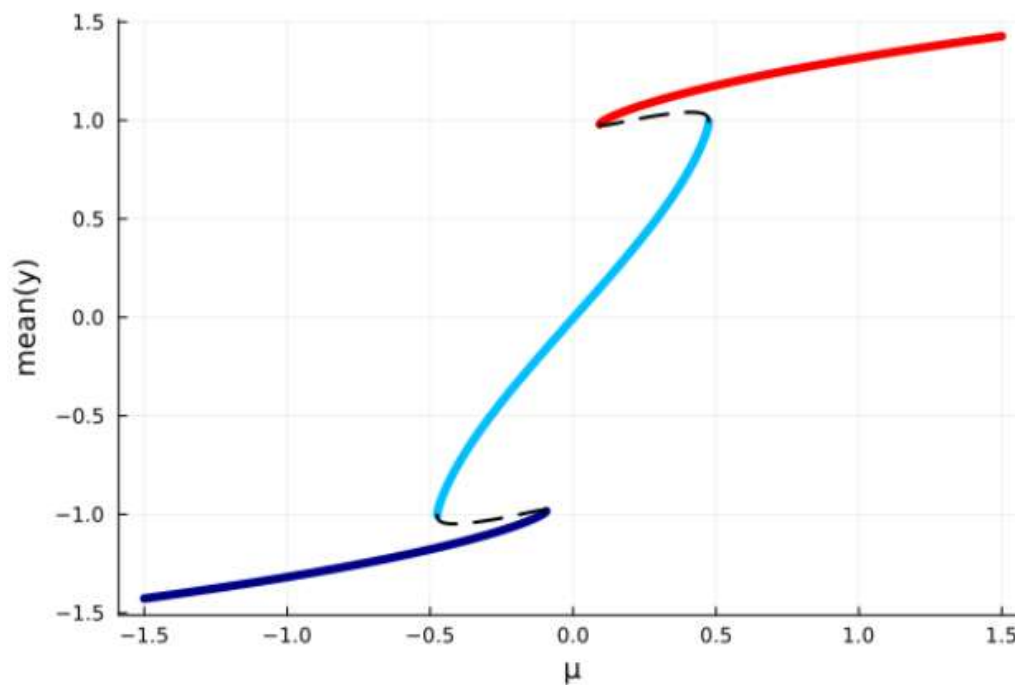
Tipping and front dynamics depend on **local** potential differences

New behaviours:

- New intermediate stable branches
- Multi-fronts can be stationary

Effect of spatial heterogeneity on bifurcation diagrams

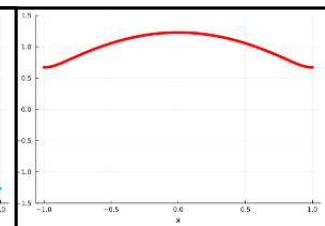
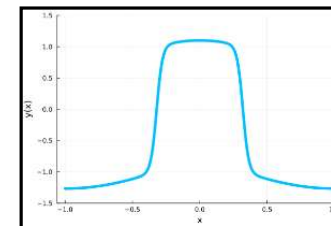
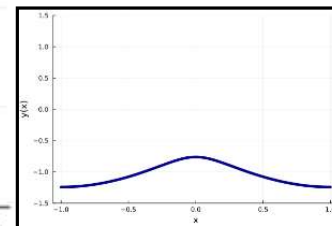
$$y_t = D y_{xx} - \partial V(y, \mathbf{x}; \mu) / \partial y$$



Tipping and front dynamics depend on **local** potential differences

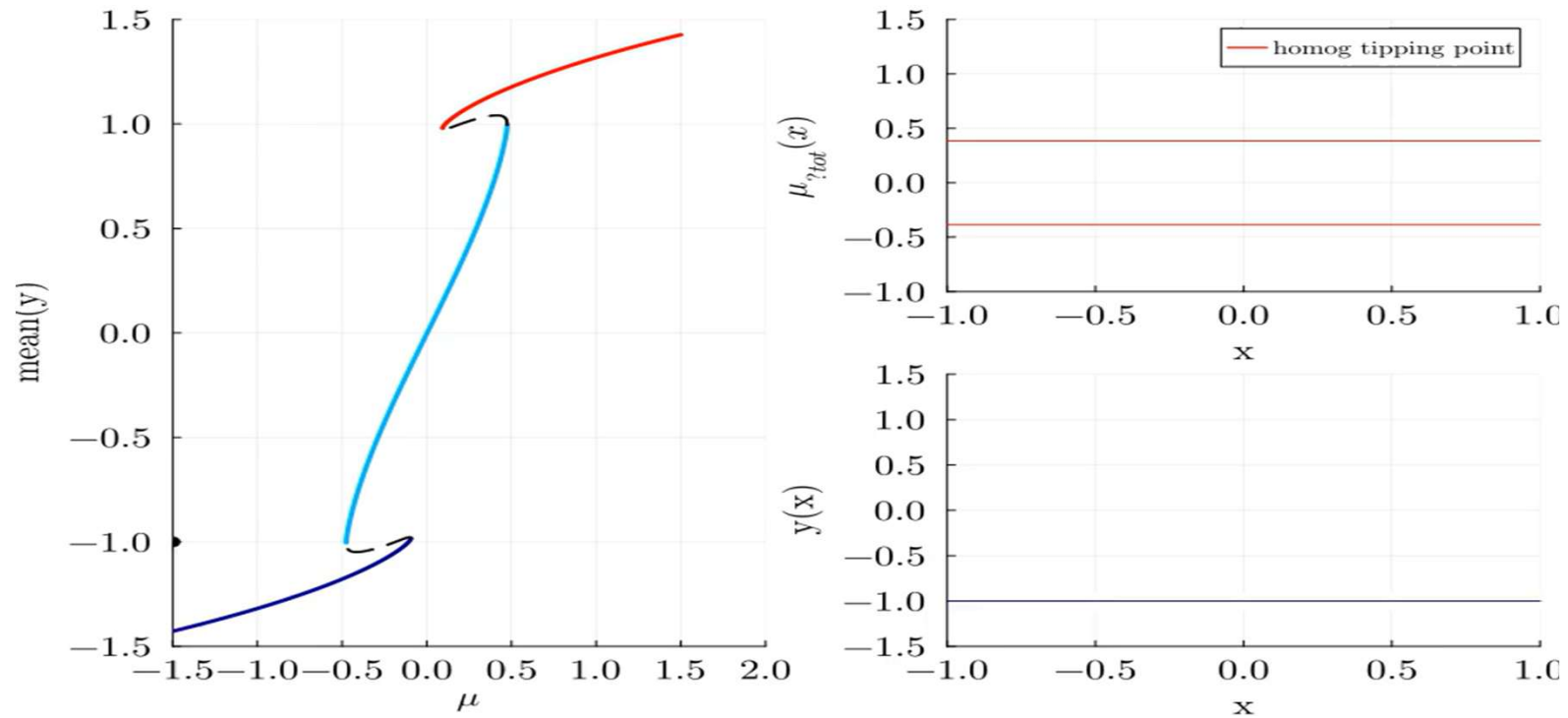
New behaviours:

- New intermediate stable branches
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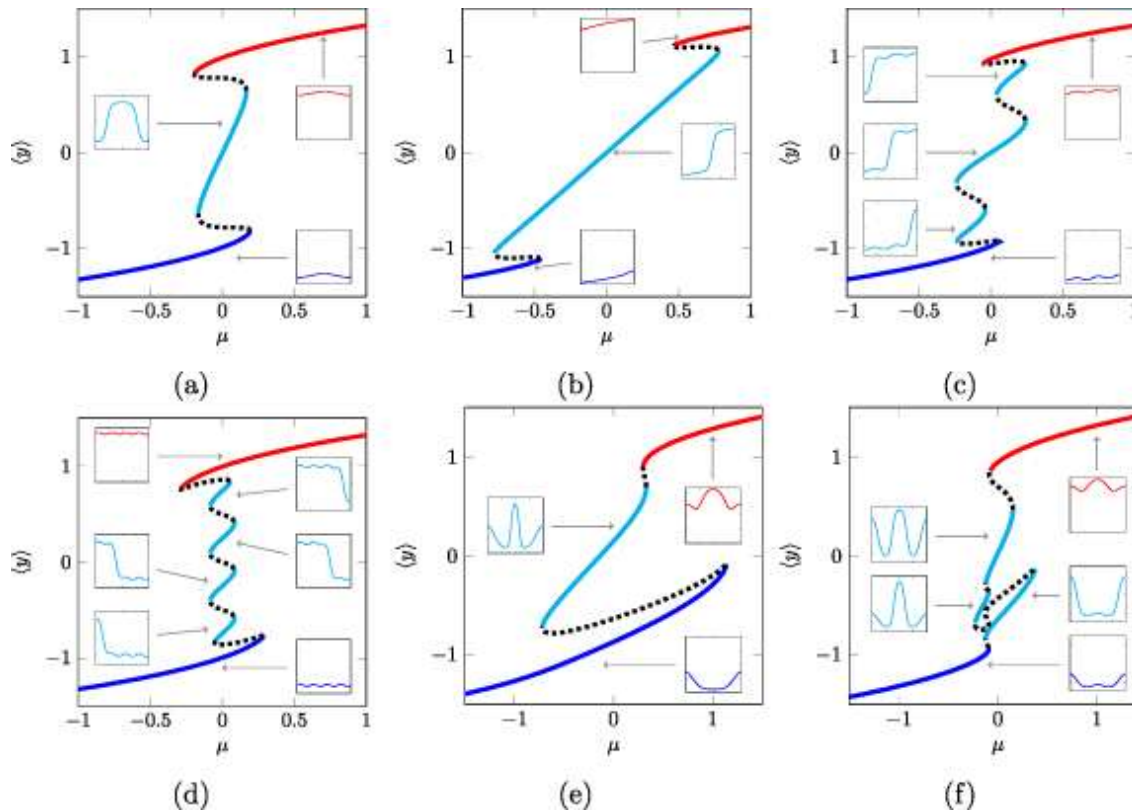


Effect of spatial heterogeneity on bifurcation diagrams

$$y_t = D y_{xx} + y(1 - y^2) + \overbrace{\mu + \cos(\pi x)}^{\mu_{tot}(x)}$$



Effect of spatial heterogeneity on bifurcation diagrams



How to choose the *best* one?

Bastiaansen, Dijkstra, Von der Heydt (2022)

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Allen-Cahn equation

Homogeneous:

$$y_t = D y_{xx} + y(1 - y^2) + \mu$$

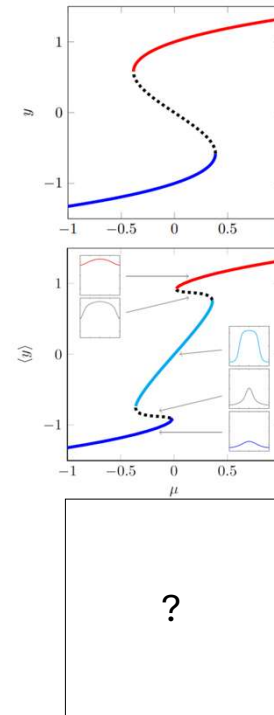
→ μ_{tip}
→ $\mu_{Maxwell}$

Heterogeneous (non optimised):

$$y_t = D y_{xx} + y(1 - y^2) + \mu + \mu_{het}(x)$$

Heterogeneous optimised:

$$y_t = D y_{xx} + y(1 - y^2) + \mu + \mu_{het}(x) + \mu_{pert}(x)$$



Roadmap

1. Spatial tipping mechanisms

- Effect of spatial heterogeneity on bifurcation diagrams
- Effect of spatial heterogeneity on front dynamics

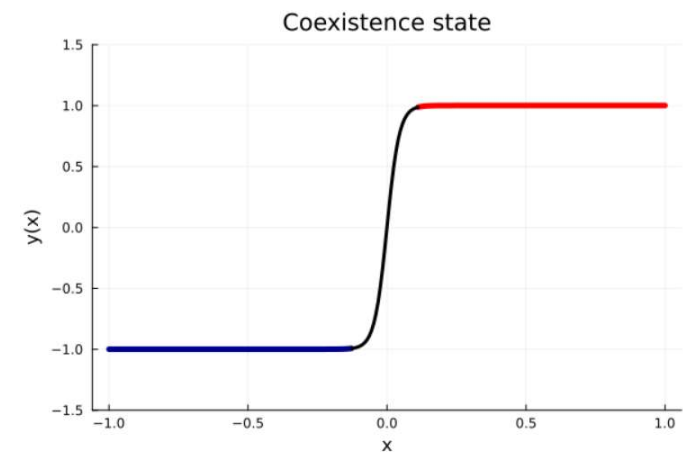
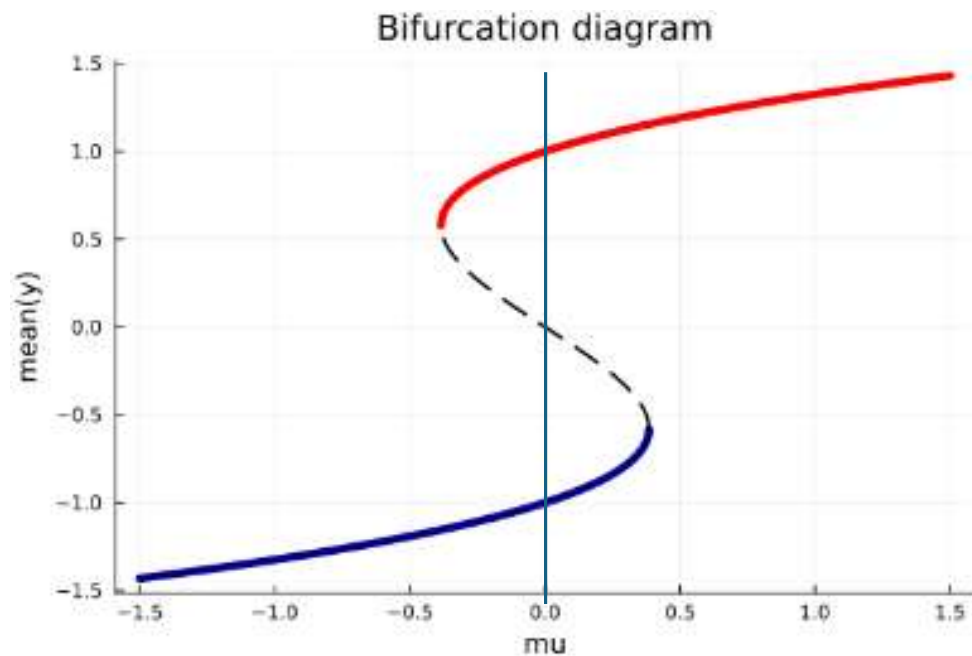
2. Optimisation framework

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Effect of spatial heterogeneity on front dynamics

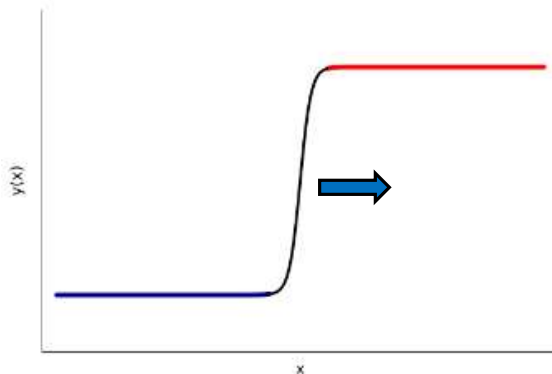
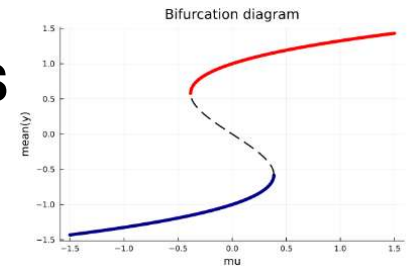
$$y_t = D y_{xx} + y(1 - y^2) + \mu$$



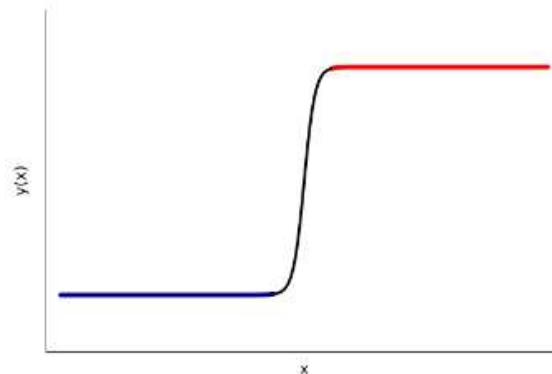
$\mu = 0$
Maxwell point

Effect of spatial heterogeneity on front dynamics

$$y_t = D y_{xx} + y(1 - y^2) + \mu$$

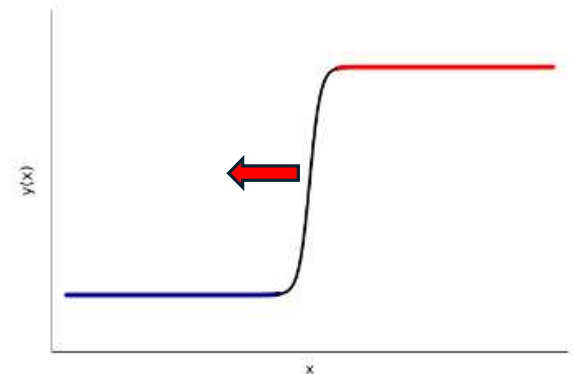


$$\mu < 0$$



$$\mu = 0$$

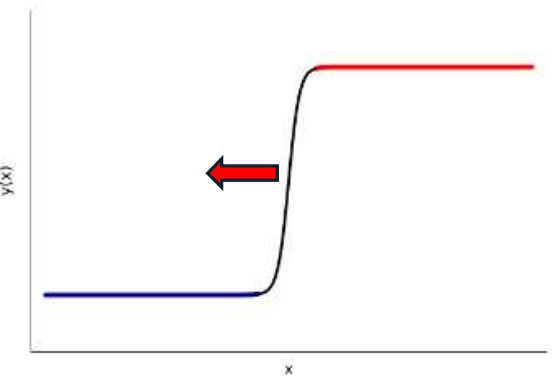
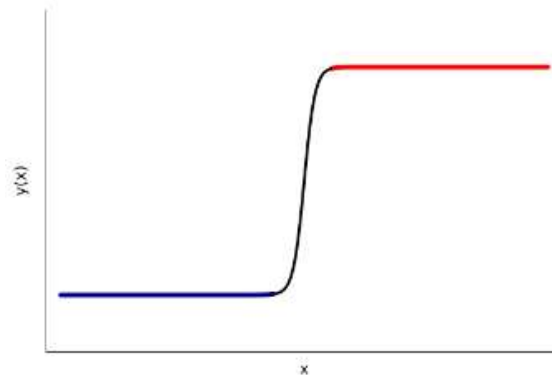
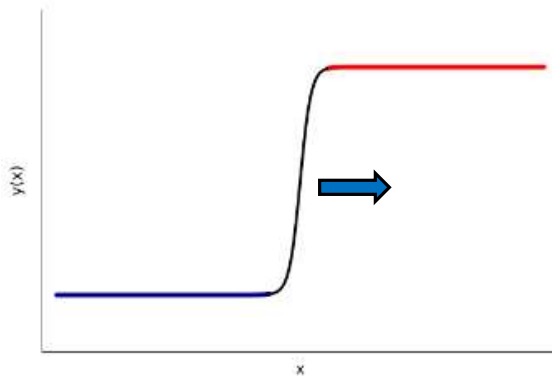
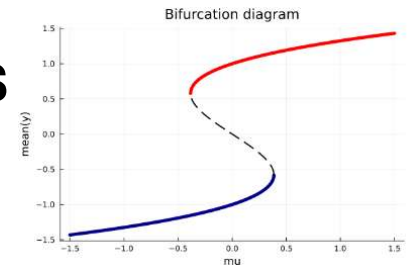
Maxwell point



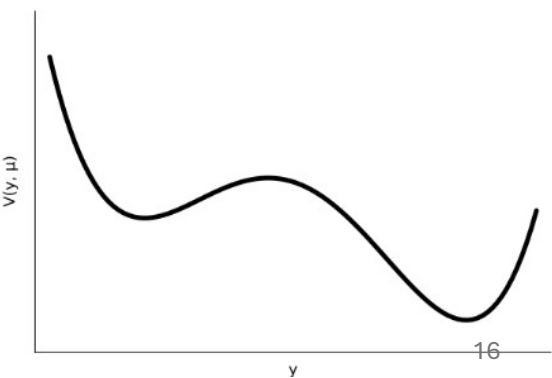
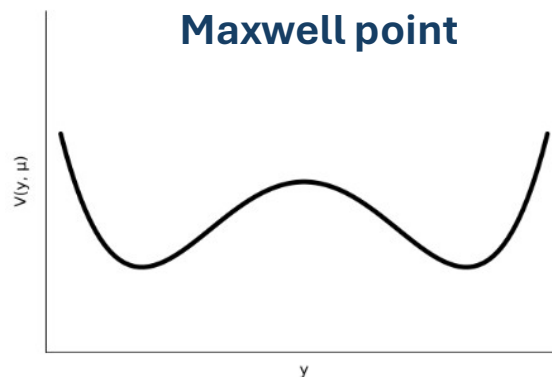
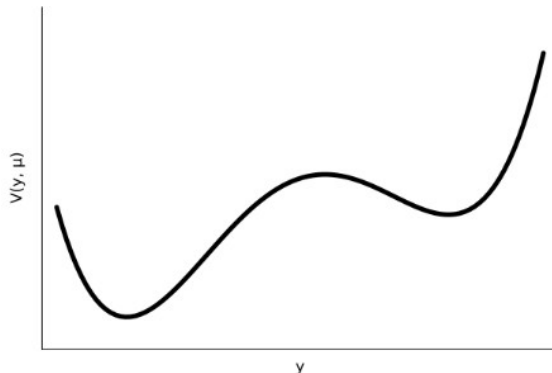
$$\mu > 0$$

Effect of spatial heterogeneity on front dynamics

$$y_t = D y_{xx} + \overbrace{y(1 - y^2) + \mu}^{-\partial V(y; \mu) / \partial y}$$

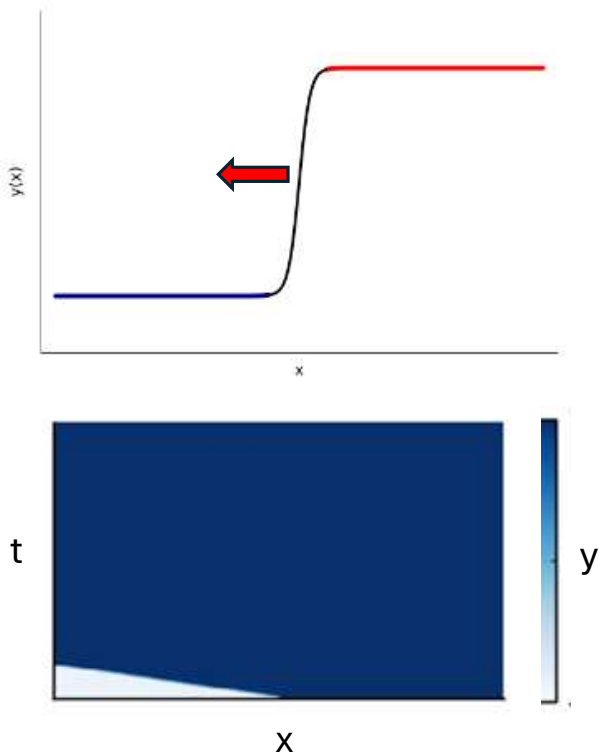


Maxwell point



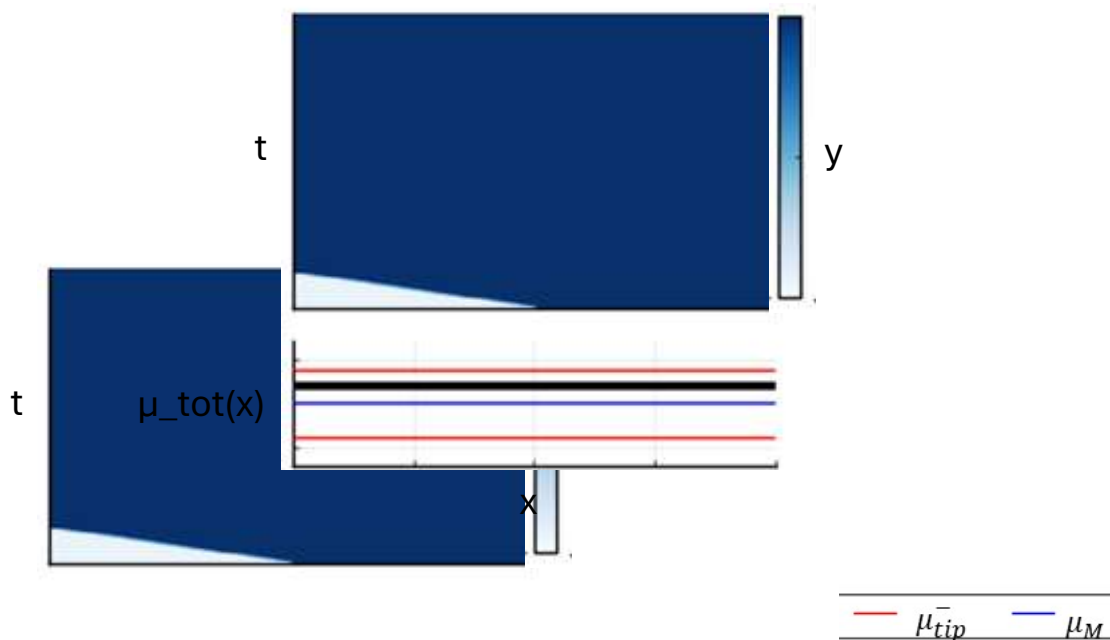
Effect of spatial heterogeneity on front dynamics

$$y_t = D y_{xx} + \frac{-\partial V(y; \mu)/\partial y}{y(1 - y^2)} + \mu$$



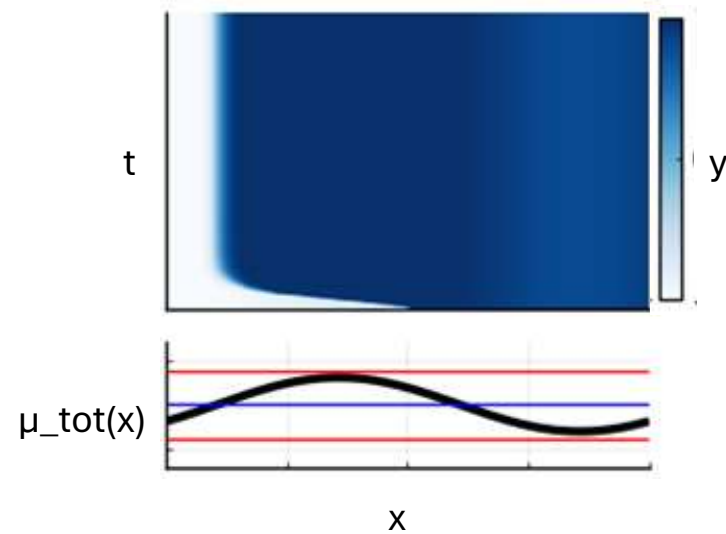
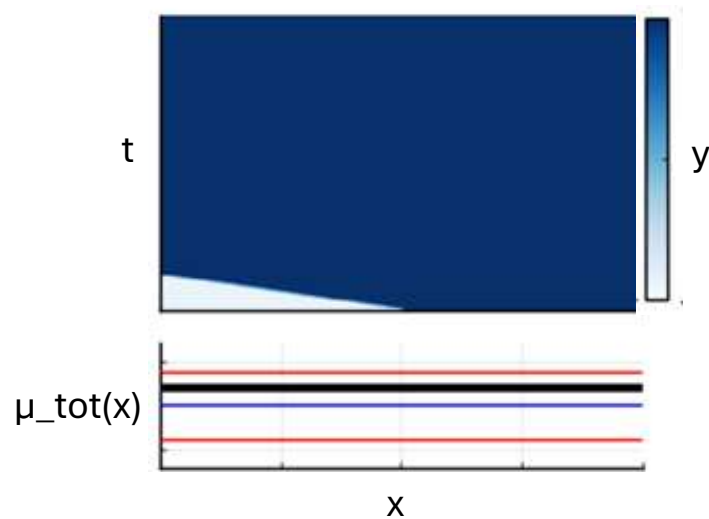
Effect of spatial heterogeneity on front dynamics

$$y_t = D y_{xx} + \overbrace{y(1 - y^2) + \mu_{tot}(x)}^{-\partial V(y, \mathbf{x}; \mu) / \partial y}$$



Effect of spatial heterogeneity on front dynamics

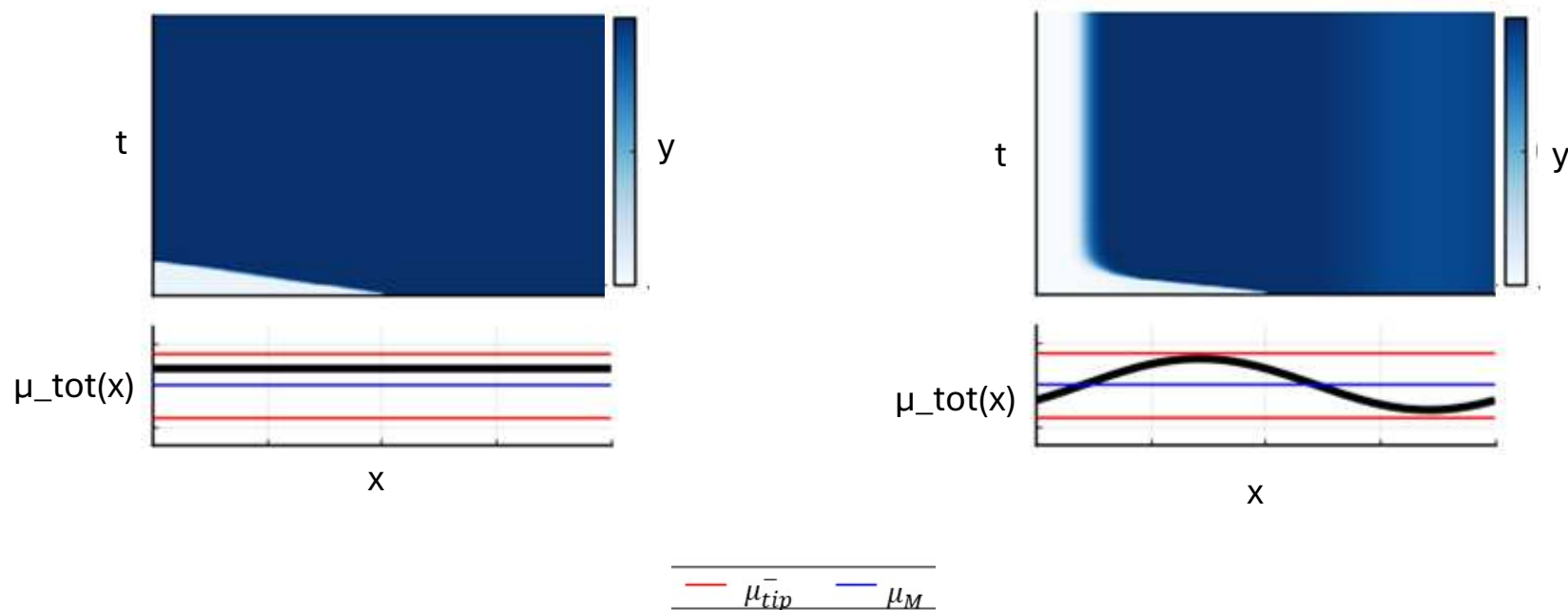
$$y_t = D y_{xx} + \overbrace{y(1 - y^2) + \mu_{tot}(x)}^{-\partial V(y, \mathbf{x}; \mu) / \partial y}$$



— μ_{tip}^- — μ_M

Effect of spatial heterogeneity on front dynamics

What is the *best* $\mu_{tot}(x)$?



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Optimisation framework

Resilience optimisation as a constrained mathematical optimisation problem

$\mu_{pert}(x)$: decision variable that represents spatial perturbations (interventions)

maximise/minimise f
 μ_{pert}

such that
$$\begin{cases} g_i(\mu_{pert}) = 0 \\ h_j(\mu_{pert}) \leq 0 \end{cases}$$

Objective function f $\leftrightarrow$ resilience proxy

Equality constraints $\leftrightarrow$ governing dynamics

Inequality constraints $\leftrightarrow$ practical limitations

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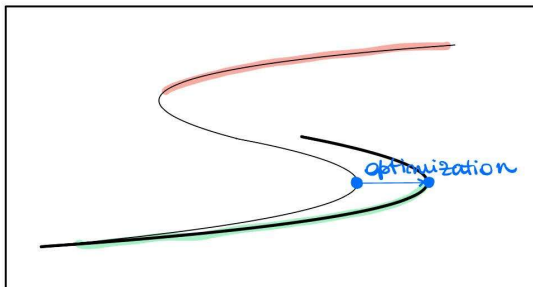
Optimisation framework

$$\begin{array}{l} \min_{\mu_{pert}} f \\ \text{such that } \begin{cases} g_i(\mu_{pert}) = 0 \\ h_j(\mu_{pert}) \leq 0 \end{cases} \end{array}$$

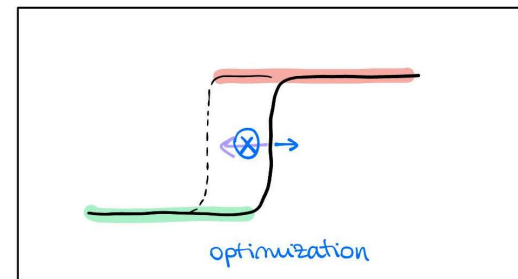
Model used: Allen-Cahn equation

$$y_t = D y_{xx} + y(1 - y^2) + \mu + \mu_{het}(x) + \mu_{pert}(x)$$

Equilibrium optimisation



Final state optimisation



Optimisation framework

$$\begin{array}{l} \min_{\mu_{pert}} f \\ \text{such that } \begin{cases} g_i(\mu_{pert}) = 0 \\ h_j(\mu_{pert}) \leq 0 \end{cases} \end{array}$$

Equilibrium optimisation

Final state optimisation

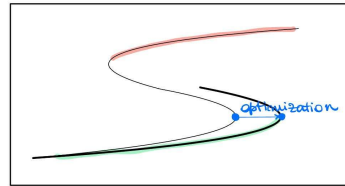
Optimisation framework

$$\min_{\mu_{pert}} f$$

such that
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Equilibrium optimisation

Objective function:
location of the saddle node



Final state optimisation

Optimisation framework

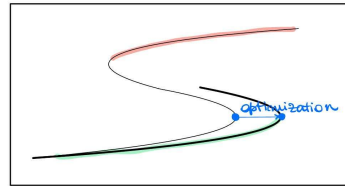
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Equilibrium optimisation

Objective function:
location of the saddle node



Equality constraint:

Bifurcation constraint:

$$D y_{xx} + y(1 - y^2) + \mu + \mu_{het}(x) + \mu_{pert}(x) = 0$$

"eigenvalue at zero"

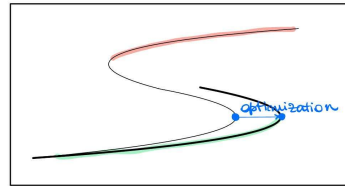
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"eigenvalue at zero"

Inequality constraint:

constraint on the perturbation: $\|\mu_{pert}(x)\| \leq \delta$

Final state optimisation

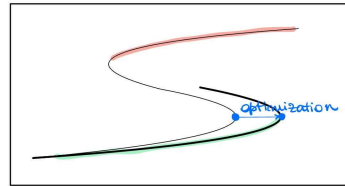
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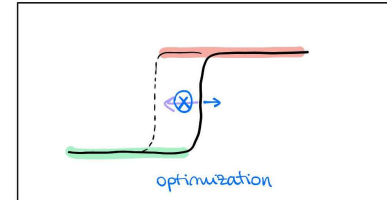
"eigenvalue at zero"

Inequality constraint:

constraint on the perturbation: $\|\mu_{pert}(x)\| \leq \delta$

Final state optimisation

Objective function:
globally averaged state at the final time



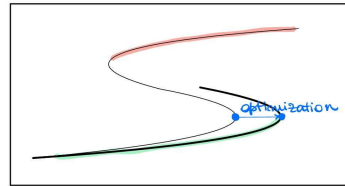
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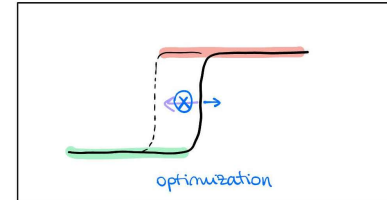
Equilibrium optimisation

Objective function:
location of the saddle node



Final state optimisation

Objective function:
globally averaged state at the final time



Equality constraint:

Bifurcation constraint:

$$D y_{xx} + y(1 - y^2) + \mu + \mu_{het}(x) + \mu_{pert}(x) = 0$$

"eigenvalue at zero"

Equality constraint:

Model specific constraint:

$$y_t = D y_{xx} + y(1 - y^2) + \mu + \mu_{het}(x) + \mu_{pert}(x)$$

Inequality constraint:

constraint on the perturbation: $\|\mu_{pert}(x)\| \leq \delta$

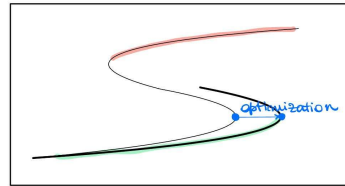
Optimisation framework

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Equilibrium optimisation

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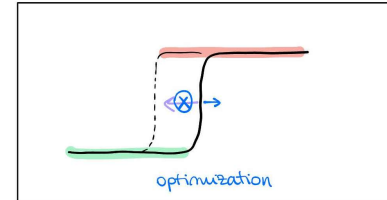
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Inequality constraint:

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Final state optimisation

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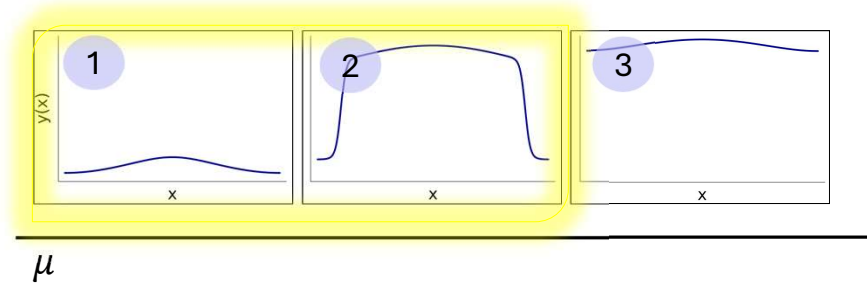
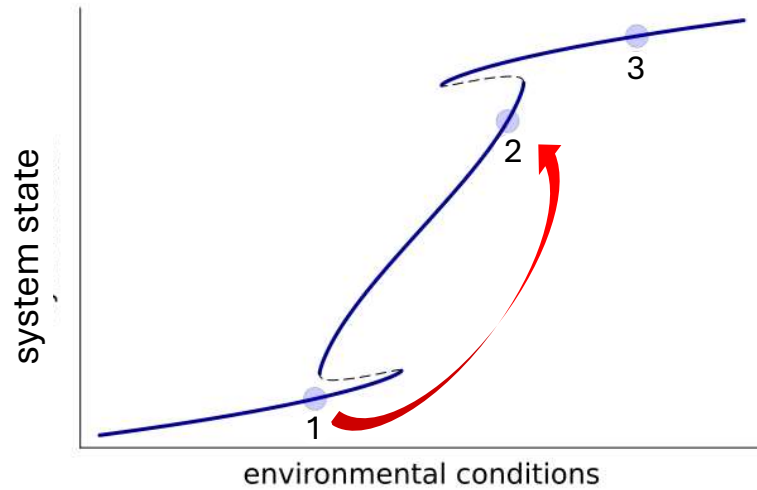
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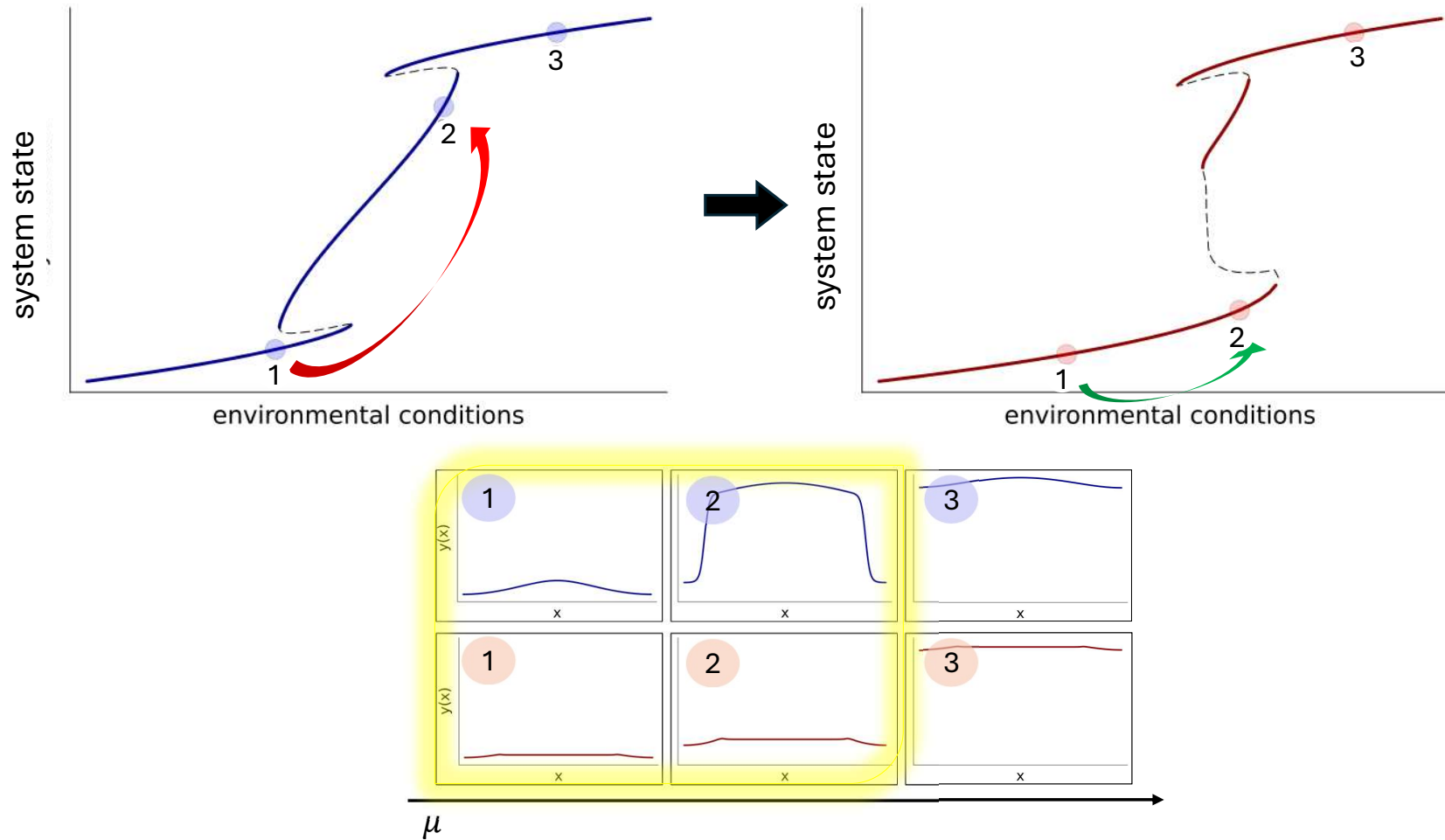
Application 1: Equilibrium optimization

Goal: optimize bifurcation diagram

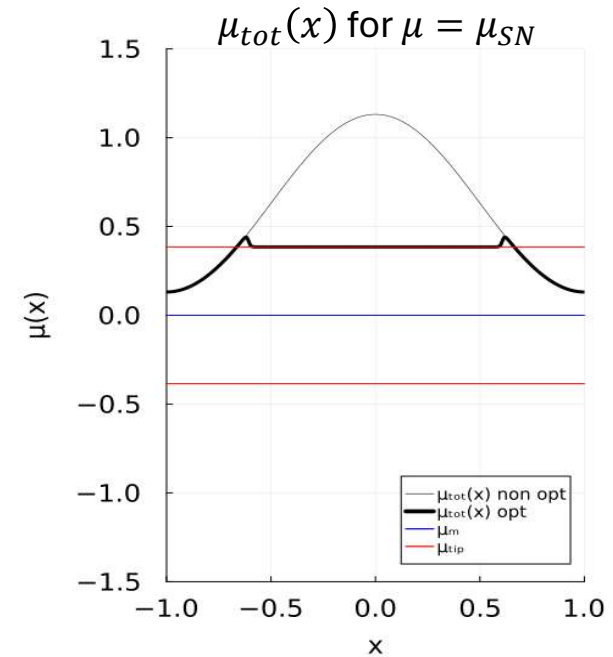
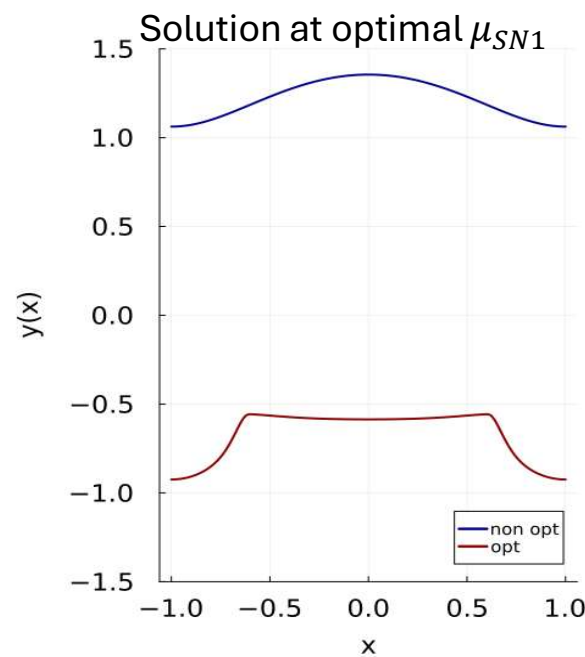
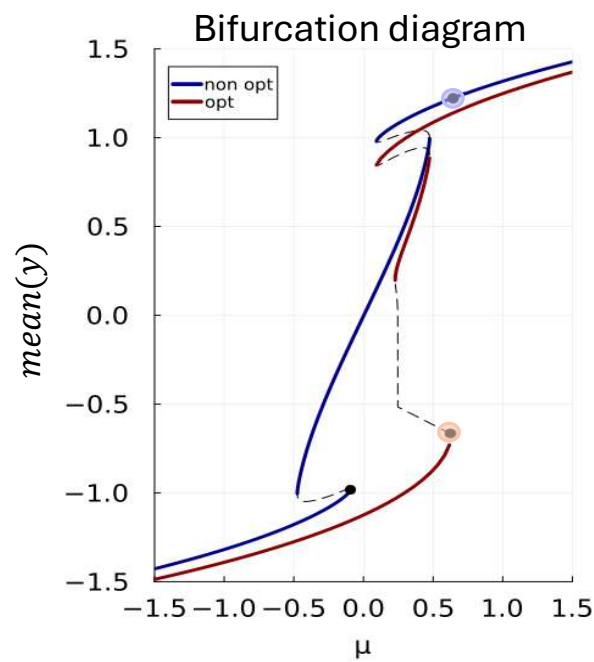
Equilibrium optimisation: example A



Equilibrium optimisation: example A



Equilibrium optimisation: example A - details

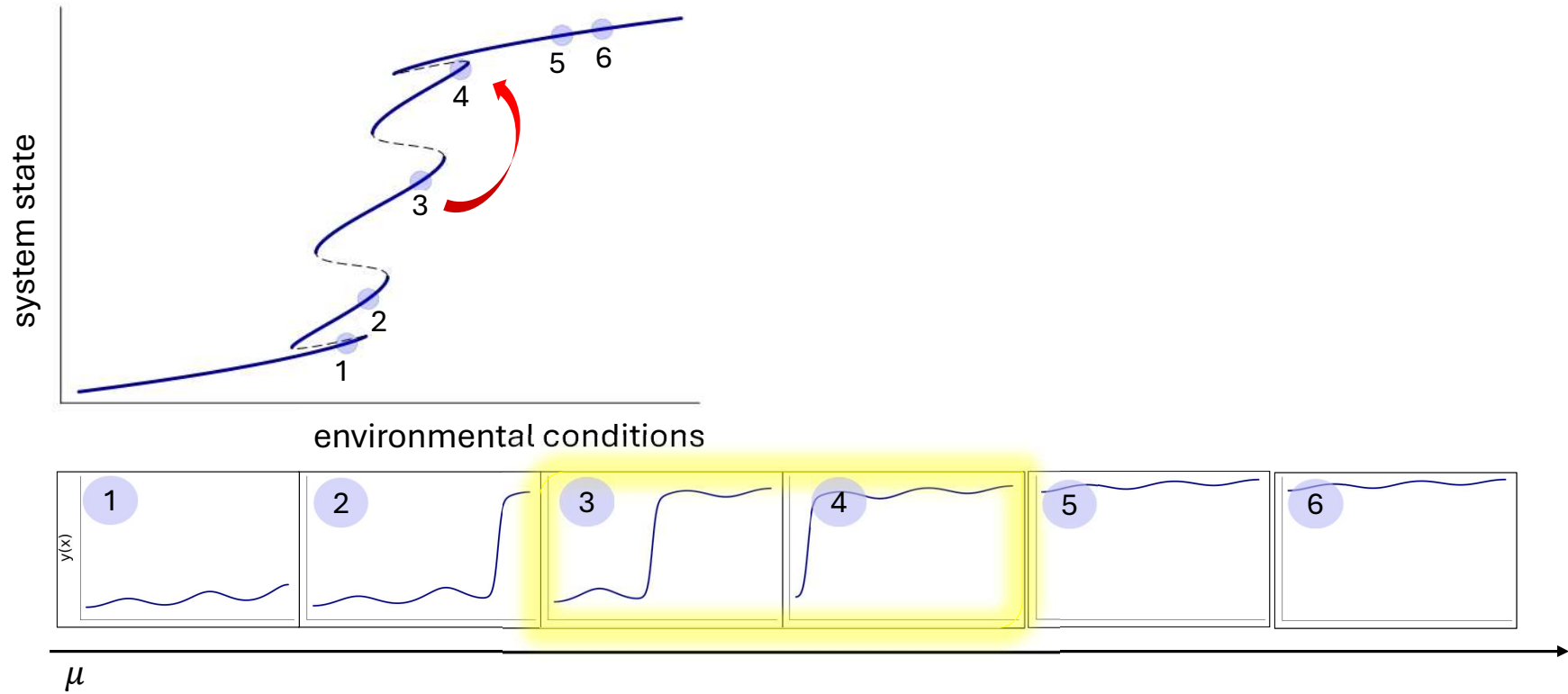


Non optimised system: $y_t = D y_{xx} + y(1 - y^2) + \mu + \frac{1}{2} \cos(\pi x)$

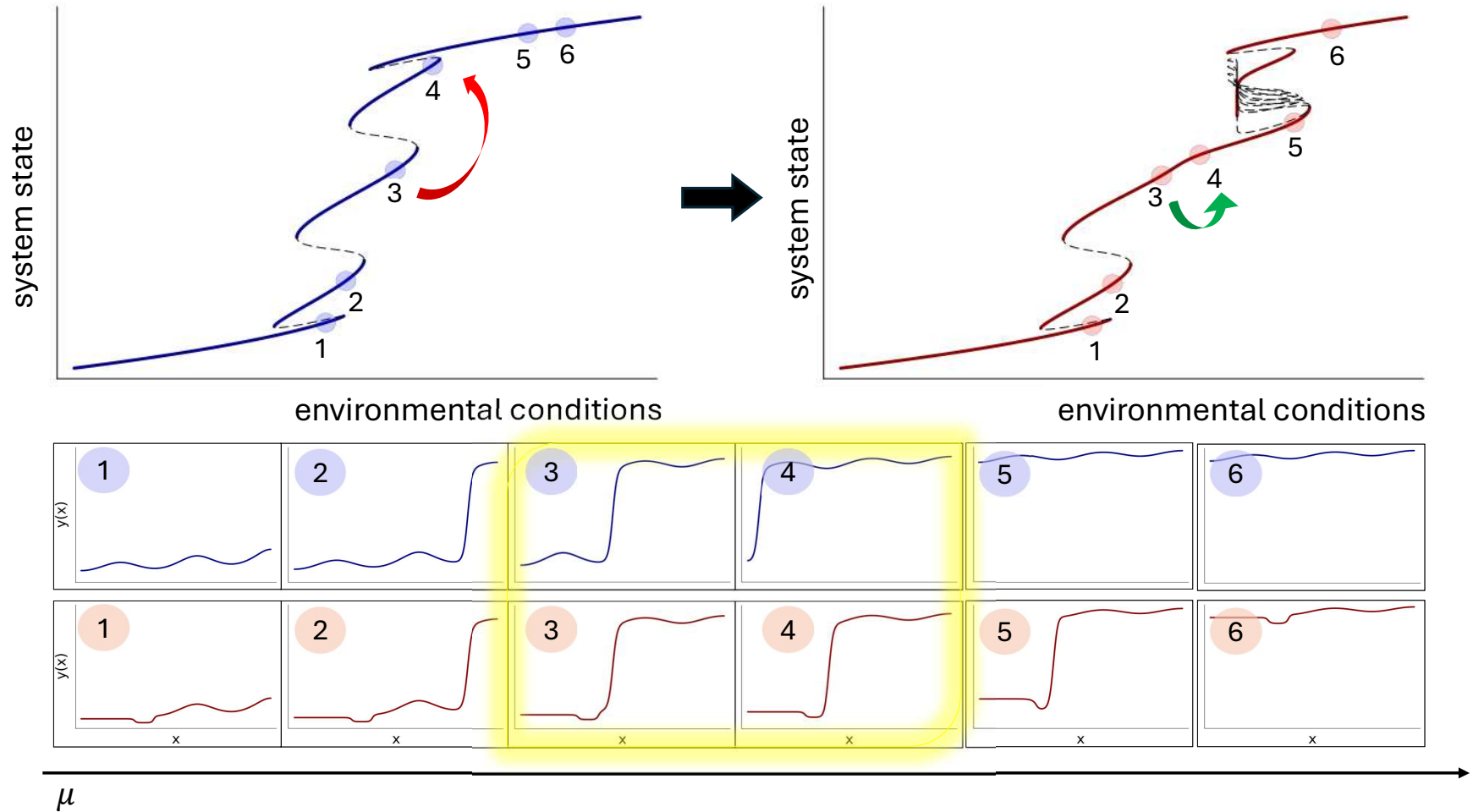
Optimised system: $y_t = D y_{xx} + y(1 - y^2) + \underbrace{\mu + \frac{1}{2} \cos(\pi x) + \mu_{pert}(x)}_{\mu_{tot}(x)}$

Objective function: $-\mu_{SN1}$

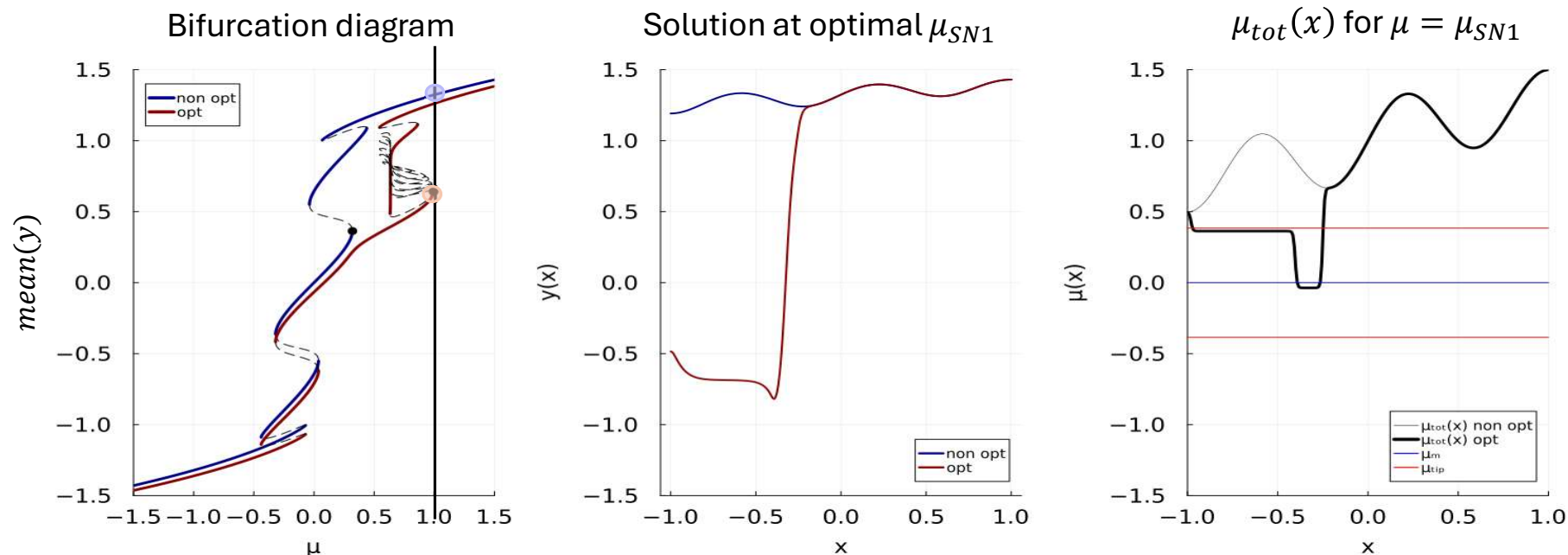
Equilibrium optimisation: example B



Equilibrium optimisation: example B



Equilibrium optimisation: example B - details



Non optimised system: $y_t = D y_{xx} + y(1 - y^2) + \mu + \frac{1}{2} \cos(\pi x) \sin(2\pi x)$ Objective function: $-\mu_{SN5}$

Optimised system: $y_t = D y_{xx} + y(1 - y^2) + \underbrace{\mu + \frac{1}{2} \cos(\pi x) \sin(2\pi x) + \mu_{pert}(x)}_{\mu_{tot}(x)}$

Application 2: Final state optimization with fixed μ

Goal: optimize final state of the simulation intervening on front movement and front creation

Interventions on front movement

Objective function:
spatial mean of y at final time t_{end}

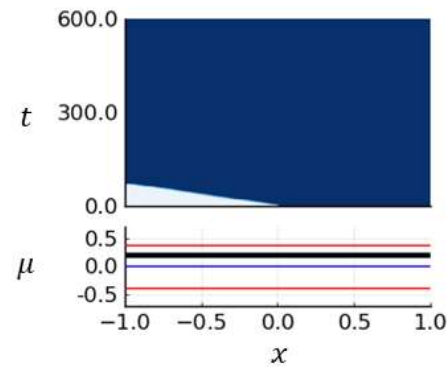
Non optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.2$$

Optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.2 + \mu_{pert}(x)$$

Non optimised system

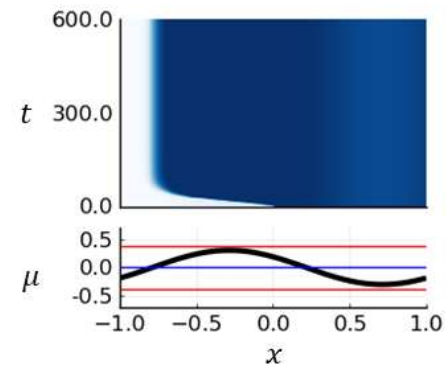


Non optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.3\cos(\pi x + 0.9)$$

Optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.3\cos(\pi x + 0.9) + \mu_{pert}(x)$$



Interventions on front movement

Objective function:
spatial mean of y at final time t_{end}

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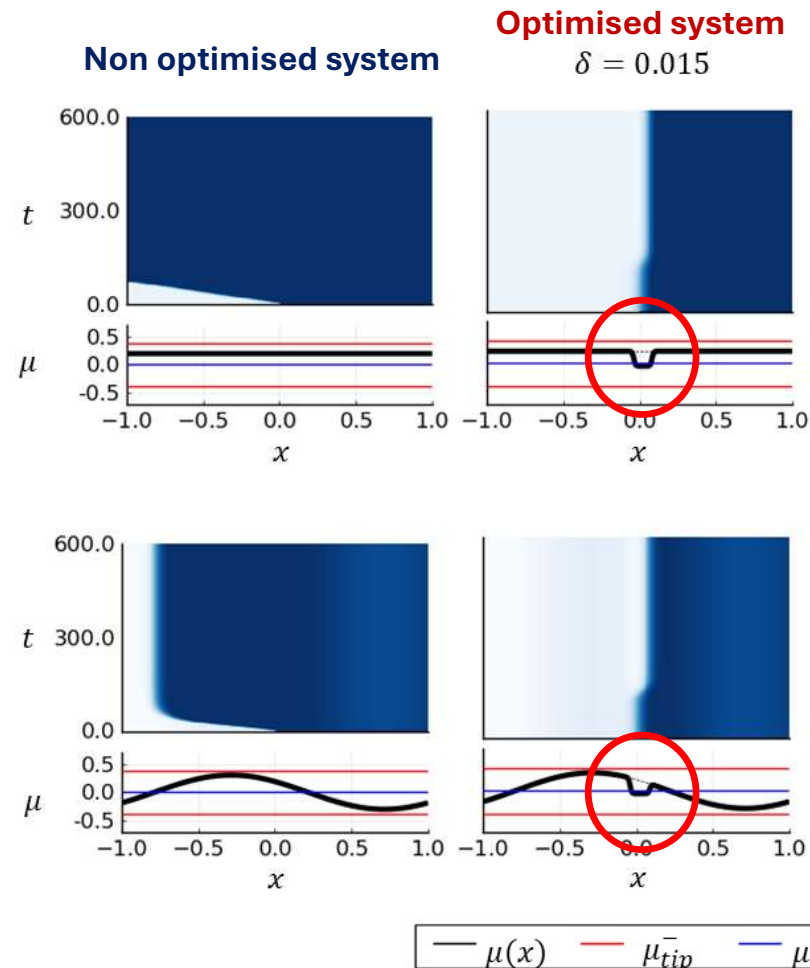
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Non optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.3\cos(\pi x + 0.9)$$

Optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.3\cos(\pi x + 0.9) + \mu_{pert}(x)$$



Interventions on front movement

Objective function:
spatial mean of y at final time t_{end}

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Optimised system:

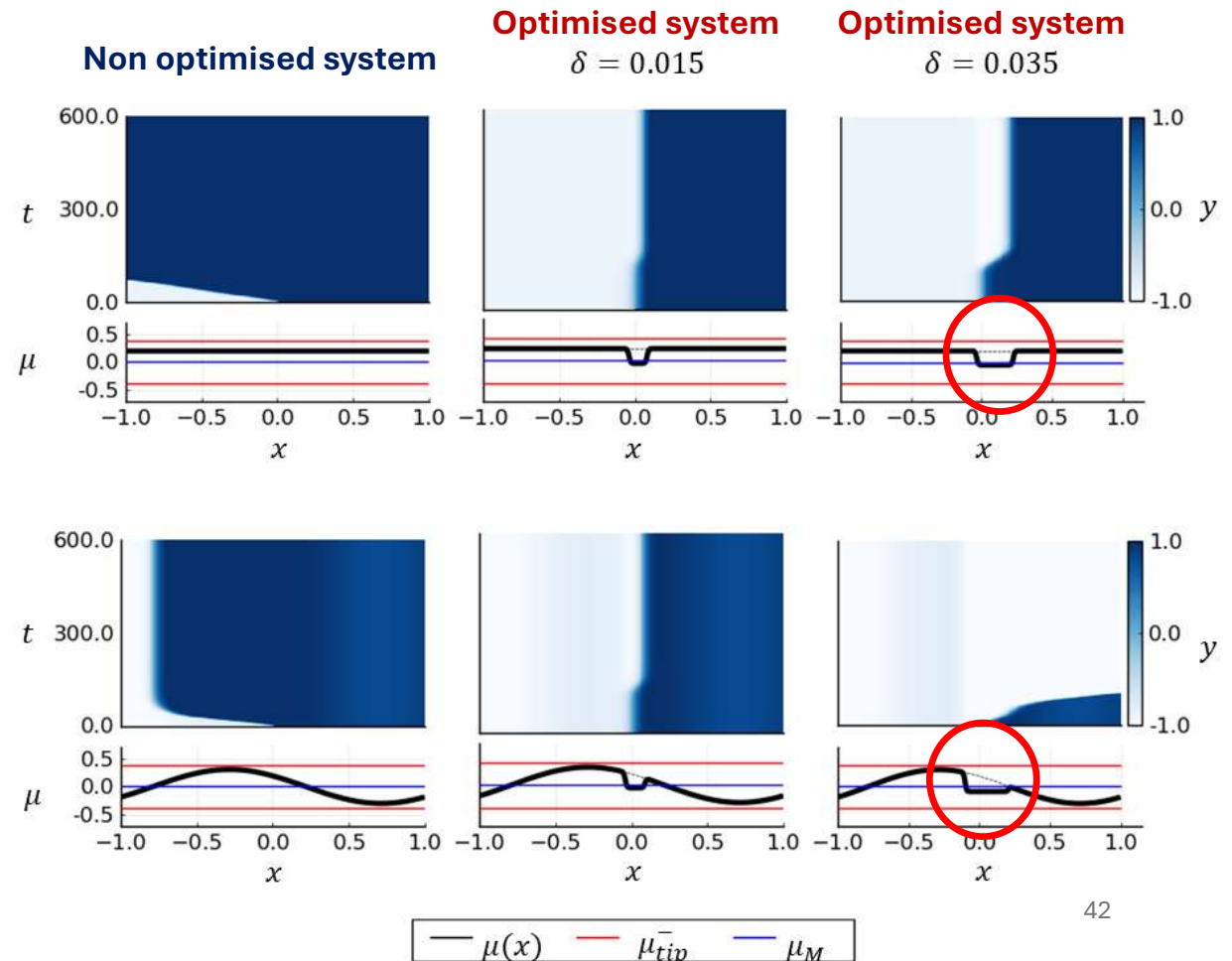
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Non optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.3\cos(\pi x + 0.9)$$

Optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.3\cos(\pi x + 0.9) + \mu_{pert}(x)$$



Interventions on front creation

Objective function:
spatial mean of y at final time t_{end}

Non optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.2$$

Optimised system:

$$y_t = D y_{xx} + y(1 - y^2) + 0.2 + \mu_{pert}(x)$$

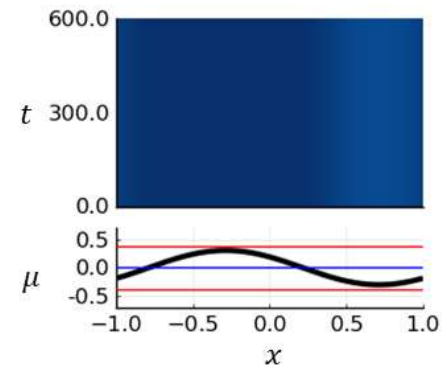
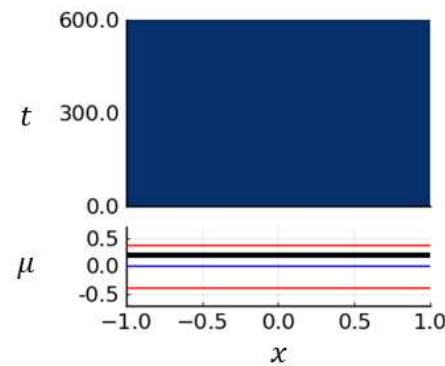
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Non optimised system



Interventions on front creation

Objective function:
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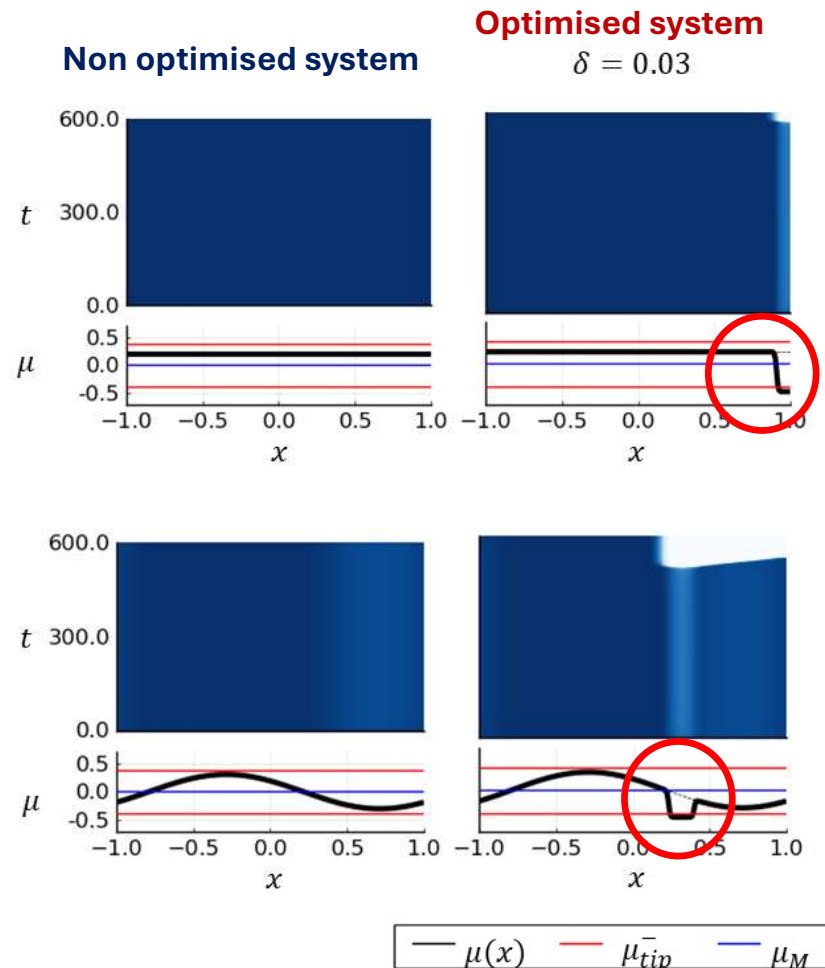
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Interventions on front creation

Objective function:
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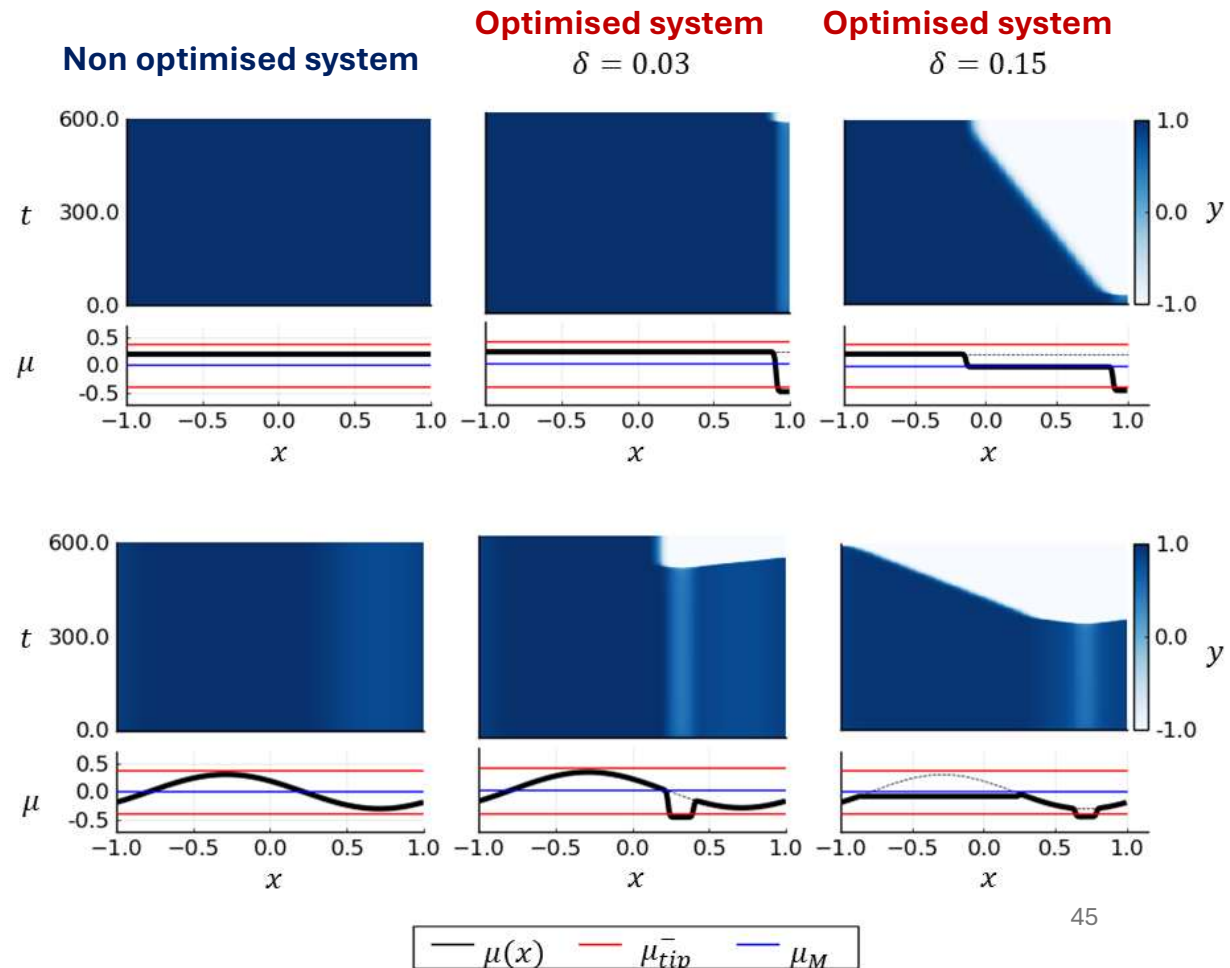
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Summary

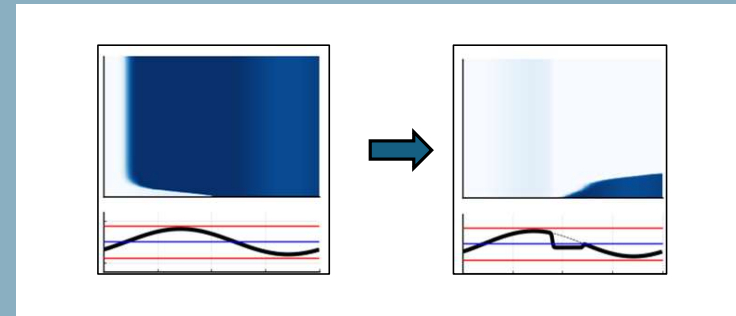
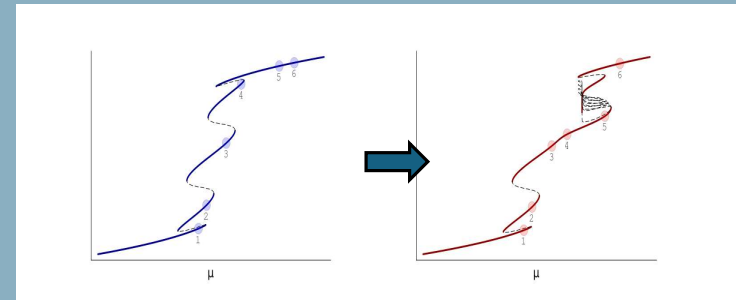
Framework

$$\min_{\mu_{pert}} f \text{ such that } \begin{cases} g_i(\mu_{pert}) = 0 \\ h_j(\mu_{pert}) \leq 0 \end{cases}$$

- Generality
- Extensibility

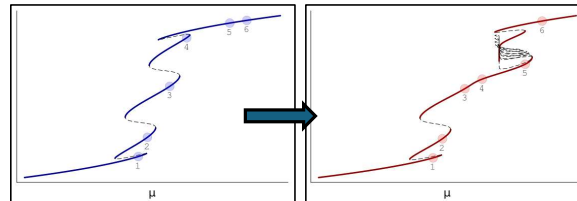
Implications

Local actions influence large-scale behaviour

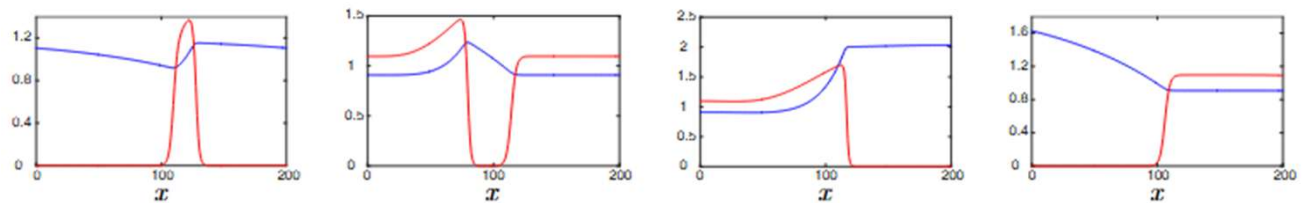


Outlook

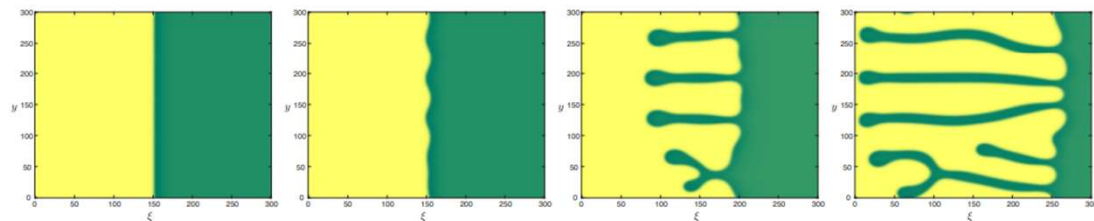
1D gradient systems
(Allen-Cahn equation)



1D non gradient systems
(extended Klausmeier models)



2D systems and front instabilities?



Numerical optimisation

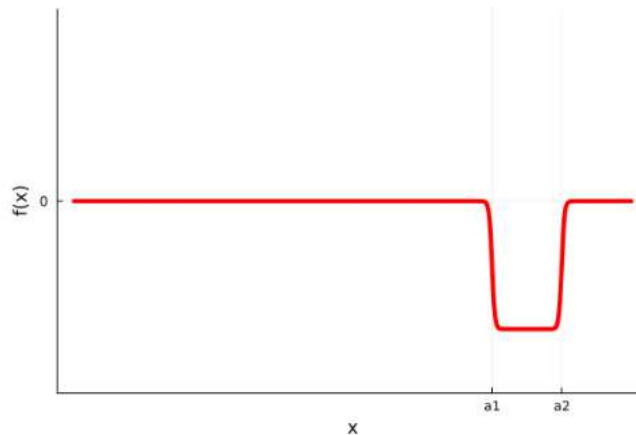
Algorithm **SLSQP** (in Julia): this is a sequential quadratic programming (SQP) algorithm for nonlinearly constrained gradient-based optimisation (supporting both inequality and equality constraints)

- Very sensitive to initial conditions
- Gradient-based

Numerical optimisation

The decision variable $\mu_{pert}(x)$ is in an infinite-dimensional space. For the optimisation, the space needs to be discretised

$$f(x; a_1, a_2) = \frac{1}{2}(1 + \tanh(\kappa(x - a_1))) \cdot \frac{1}{2}(1 - \tanh(\kappa(x - a_2)))$$

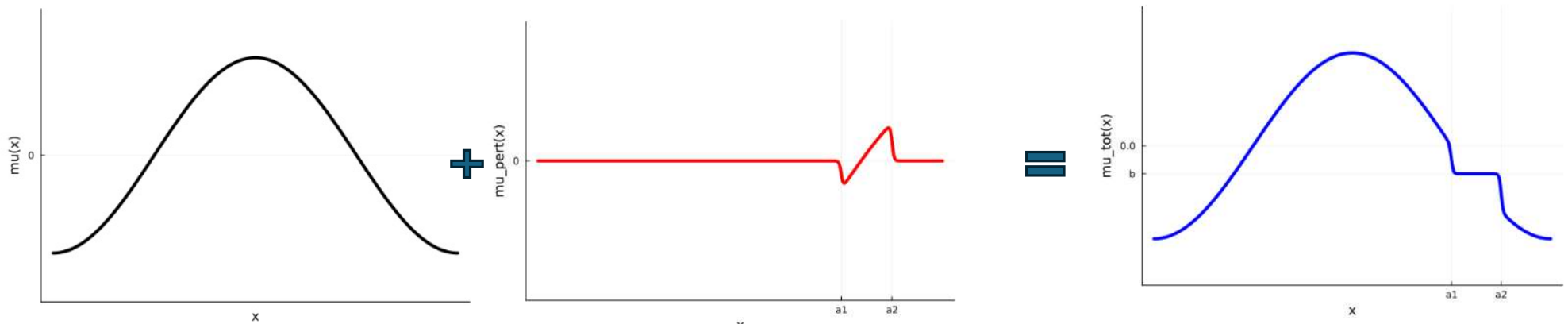


Aurora Faure Ragani (a.faureragani@uu.nl)

Numerical optimisation

The decision variable $\mu_{pert}(x)$ is in an infinite-dimensional space. For the optimisation, the space needs to be discretised!

$$\bar{f}(x; a_1, a_2, b) = f(x; a_1, a_2) \cdot (\mu(x) - b) \longrightarrow \mu_{tot}(x) = \mu(x) + \bar{f}(x; a_1, a_2, b)$$

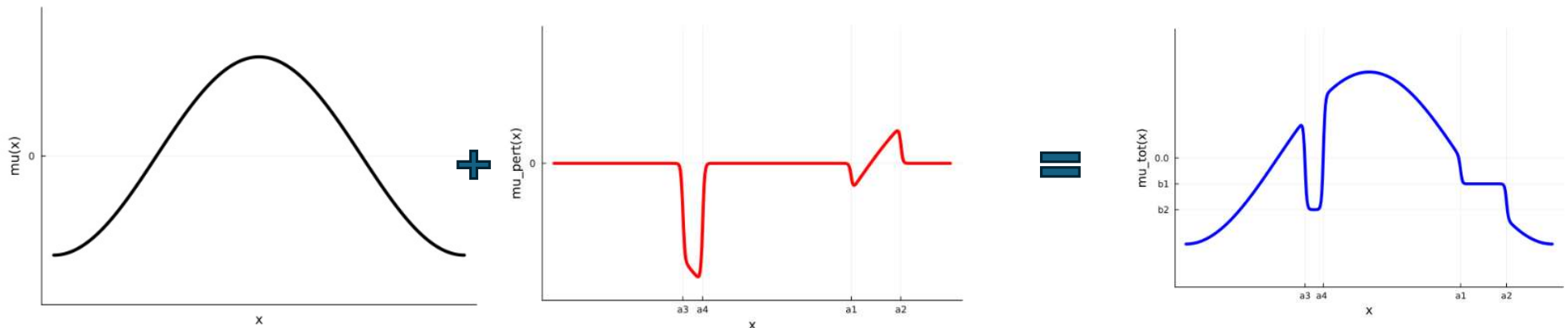


Aurora Faure Ragani (a.faureragani@uu.nl)

Numerical optimisation

The decision variable $\mu_{pert}(x)$ is in an infinite-dimensional space. For the optimisation, the space needs to be discretised!

$$\mu_{pert}(x) = \sum_{2i} \bar{f}(x; a_i, a_{i+1}, b_i) \longrightarrow \mu_{tot}(x) = \mu(x) + \mu_{pert}(x)$$



Aurora Faure Ragani (a.faureragani@uu.nl)

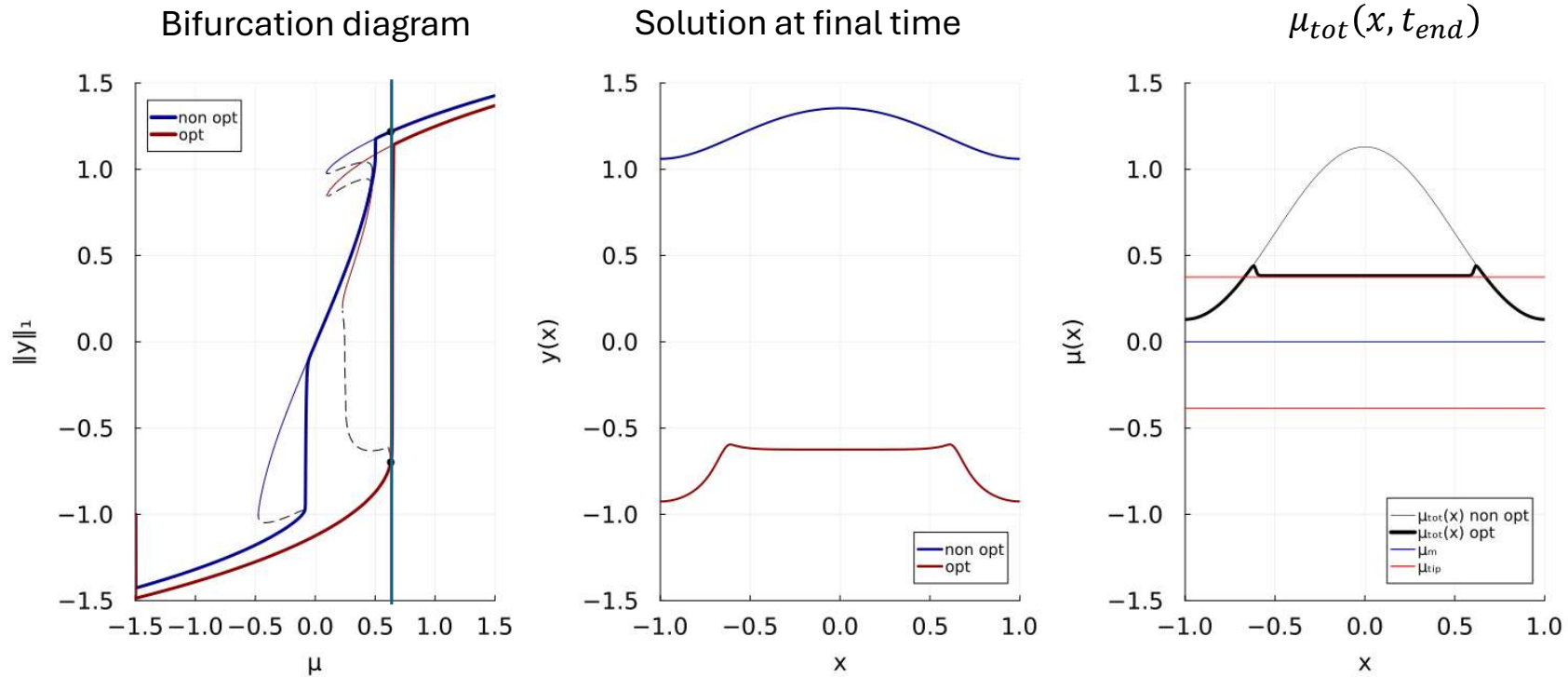
Application 3:

Final state optimization along equilibria

Goal: optimize final state of the simulation.

Simulation has slowly increasing μ and stops at μ_{end}

Extra: final state optimisation along equilibria



Non optimised system: $y_t = D y_{xx} + y(1 - y^2) + \mu(t) + \frac{1}{2} \cos(\pi x)$

Optimised system: $y_t = D y_{xx} + y(1 - y^2) + \underbrace{\mu(t) + \frac{1}{2} \cos(\pi x) + \mu_{pert}(x)}_{\mu_{tot}(x, t)}$

Objective function:

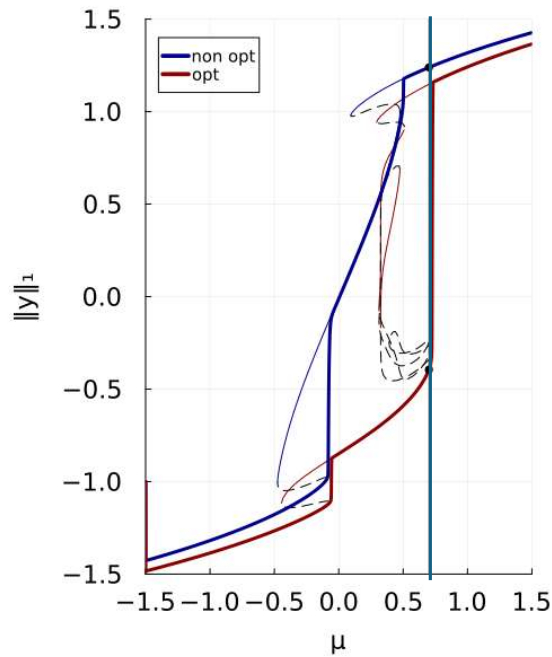
$\langle y(x, t_{end}; \mu) \rangle_x$

t_{end} s.t.

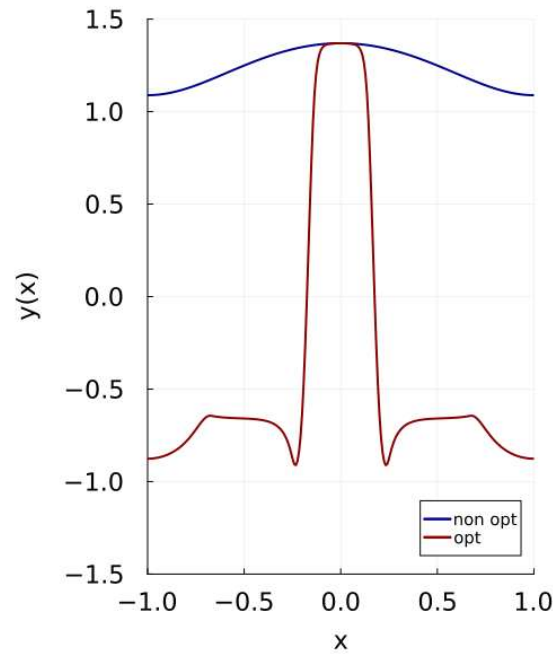
$\mu_{end} = \mu_{SN1,opt}$

Extra: final state optimisation along equilibria

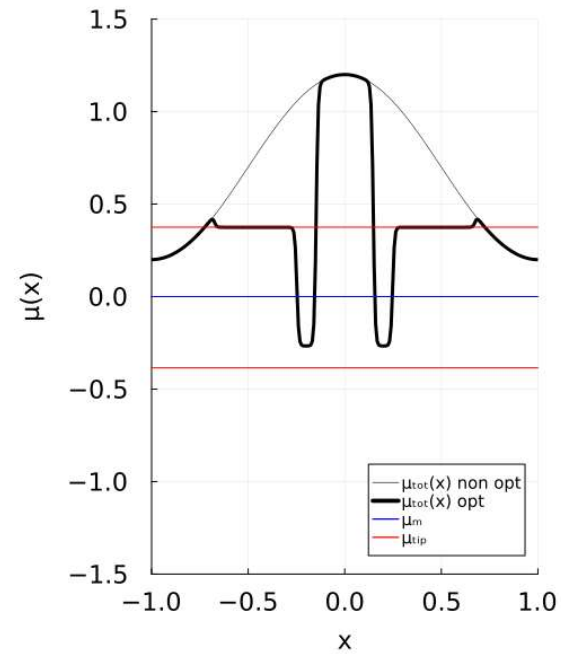
Bifurcation diagram



Solution at final time



$\mu_{tot}(x, t_{end})$



Non optimised system: $y_t = D y_{xx} + y(1 - y^2) + \mu(t) + \frac{1}{2} \cos(\pi x)$

Optimised system: $y_t = D y_{xx} + y(1 - y^2) + \underbrace{\mu(t) + \frac{1}{2} \cos(\pi x) + \mu_{pert}(x)}_{\mu_{tot}(x, t)}$

Objective function:

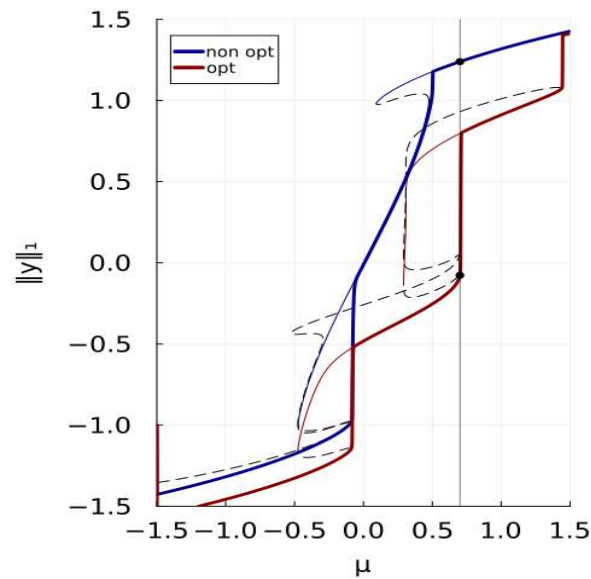
$\langle y(x, t_{end}; \mu) \rangle_x$

t_{end} s.t.

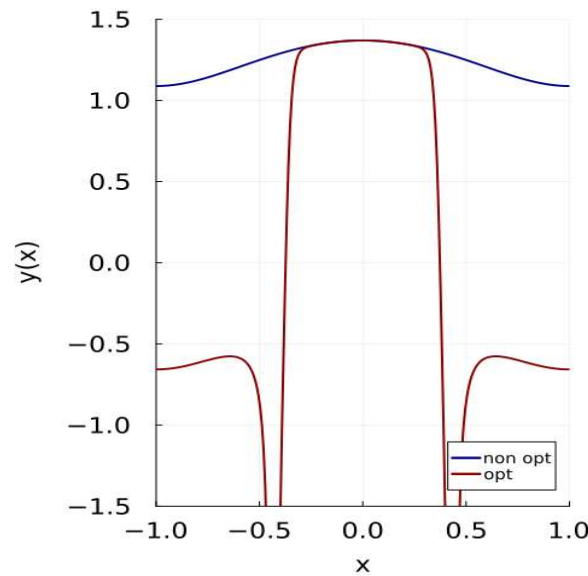
$\mu_{end} > \mu_{SN1, opt}$

Extra: example of non-additive perturbation

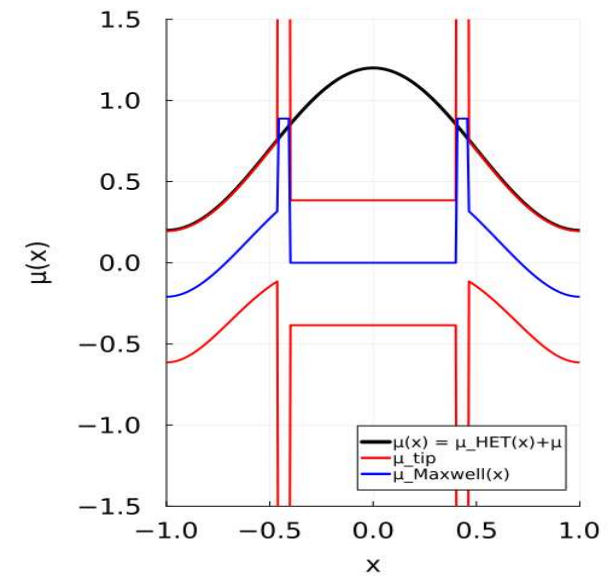
Bifurcation diagram



Solution at final time



$\mu_{tot}(x, t_{end})$
optimised case



Non optimised system: $y_t = D y_{xx} + y(1 - y^2)$

Optimised system: $y_t = D y_{xx} + (y + f_{pert}(x))(1 - y^2) + \underbrace{\mu(t) + \frac{1}{2} \cos(\pi x)}_{\mu_{tot}(x, t)}$

Objective function:
 $\langle y(x, t_{end}; \mu) \rangle_x$